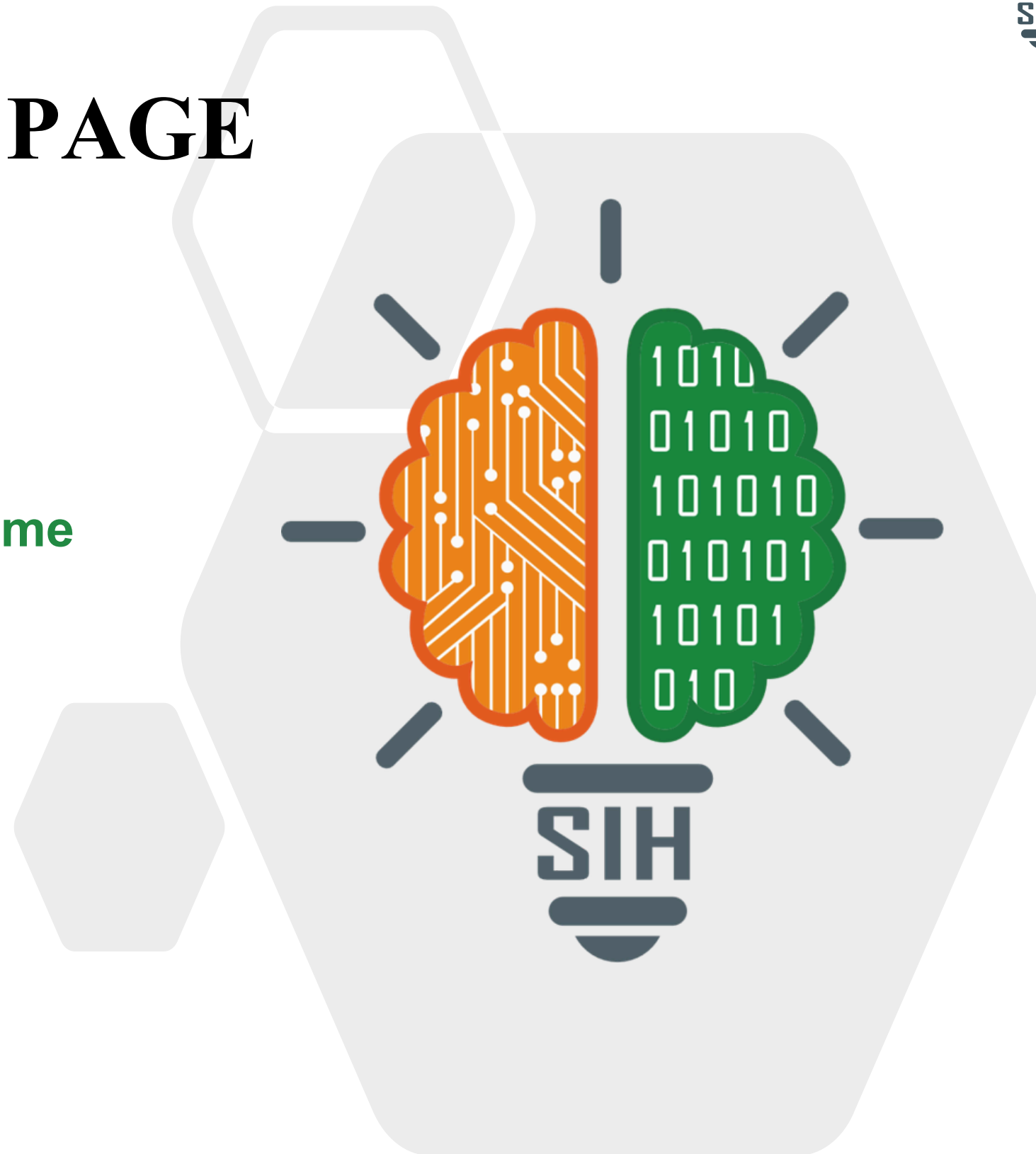


TITLE PAGE

- Problem Statement ID – 26092
- Problem Statement Title – AI-Driven Scheme
Matching for Marginalized Entrepreneurs
- Theme – Smart Automation
- PS Category – Software
- Team ID – 118918
- Team Name – AthenaX



Problem:

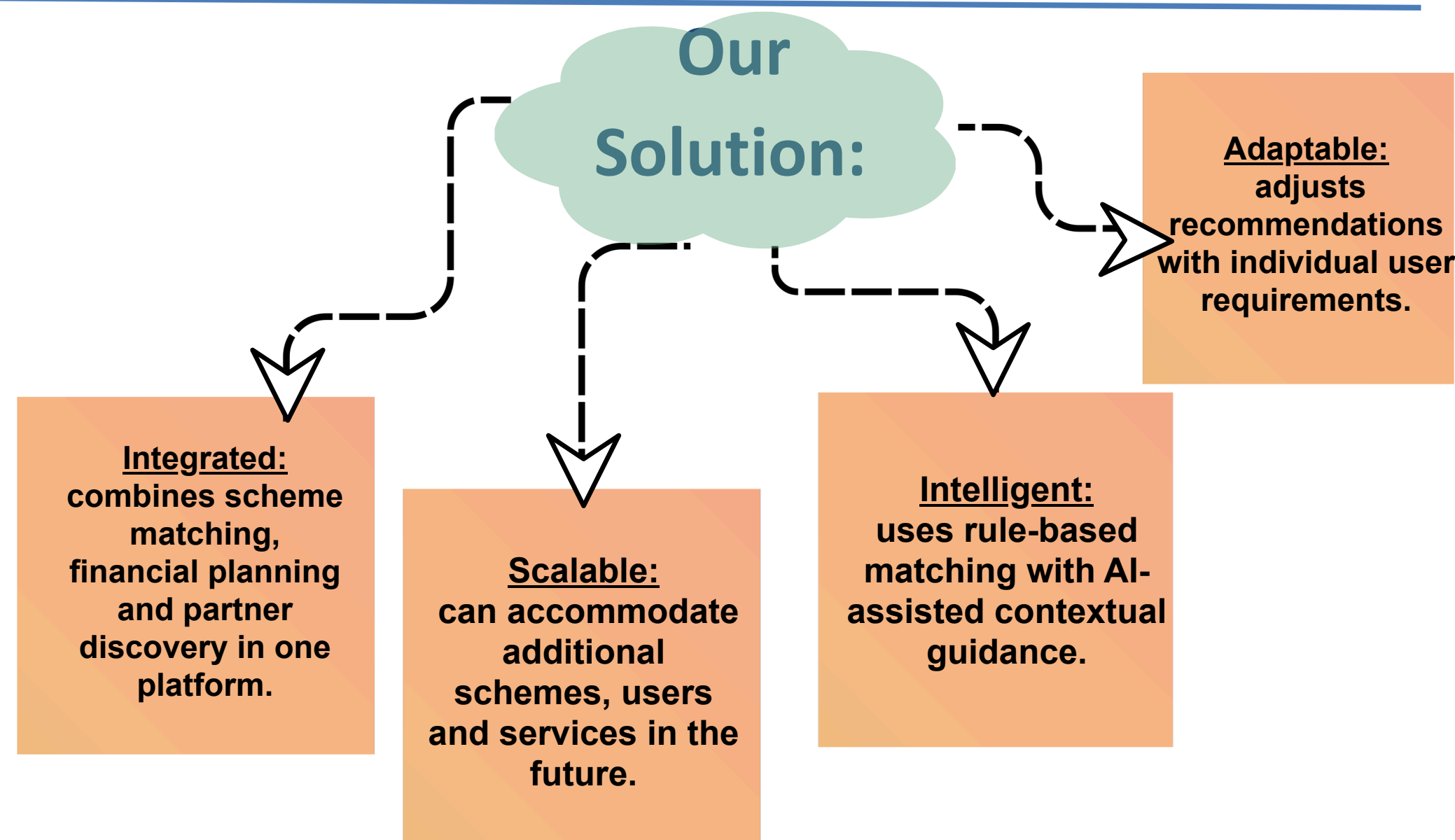
- Difficulty **identifying** the **right government loan scheme** due to varying eligibility, loan limits and financial conditions. **Lack of awareness** about the relevant Channel Partner causes **delays** and unnecessary visits.

Uniqueness:

- It **transforms complex government scheme** information into a clear and an **actionable decision** like which scheme to choose, how to plan repayment, and where to apply **all in one platform**.

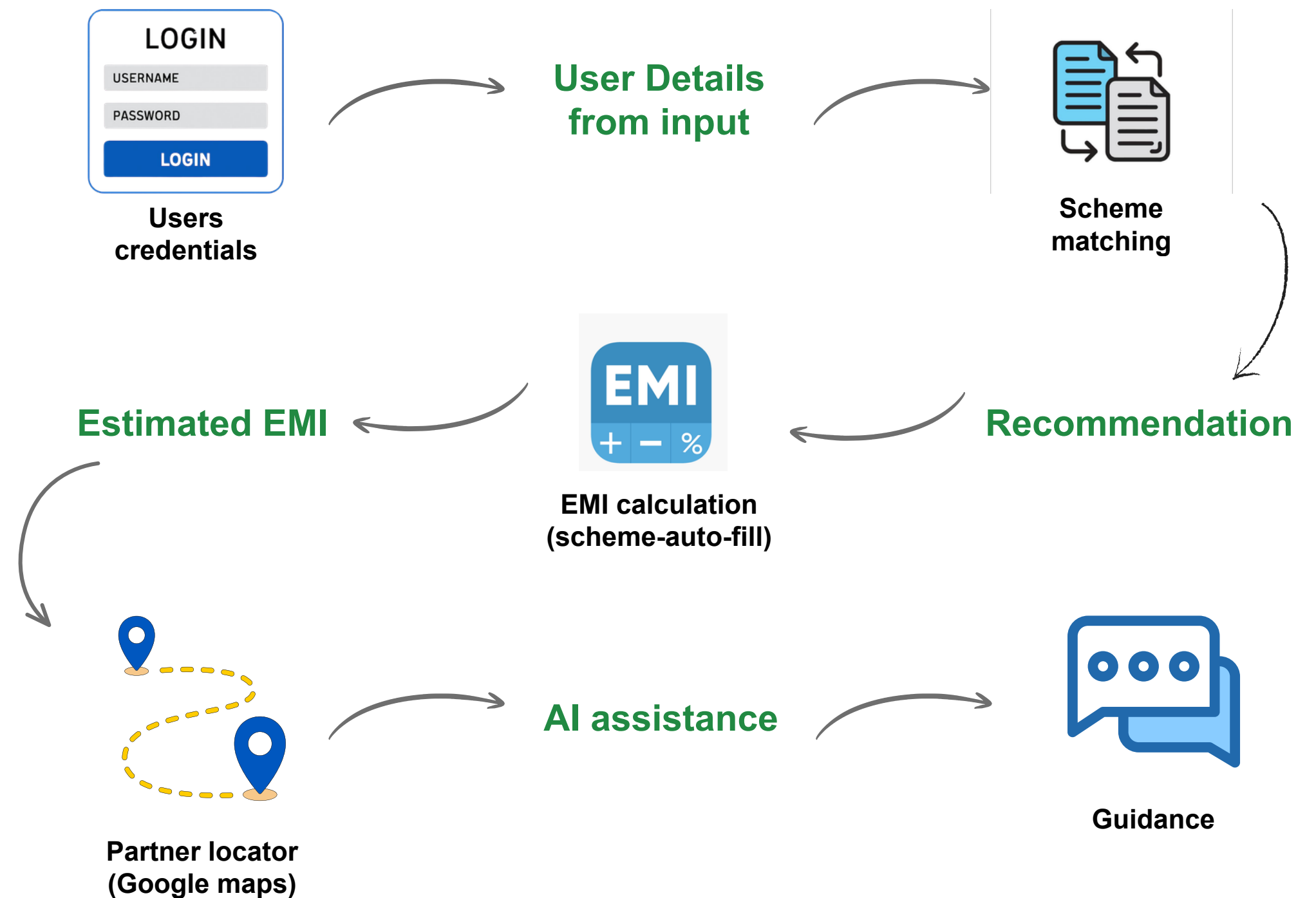
Proposed Solution :

- Rule-based scheme matching** analyzes user parameters and recommends the most suitable scheme.
- EMI calculator** estimates monthly repayment using loan amount, interest rate, tenure and moratorium period.
- Location-based Channel Partner Locator** helps users find relevant nearby application centers.
- AI-assisted contextual chatbot** provides guidance for scheme-related queries and application confusion.



Software:

- **Frontend:** Kotlin + Jetpack Compose — Android mobile application
- **Backend:** Firebase — user data and application data management
- **AI:** Google Gemini API — contextual reasoning and scheme-related assistance
- **Matching Engine:** Rule-based eligibility engine — scheme recommendation based on user parameters
- **Location Services:** Google Maps API — Channel Partner location and navigation
- **Database:** Firebase Firestore — scheme and Channel Partner data
- **Languages:** English + Hindi for improved accessibility



Feasibility

- Scheme Recommender and EMI Calculator use **well-defined numeric rules** from **official NSFDC guidelines**.
- Fully buildable end-to-end with modern, production-grade stack: **Kotlin, Jetpack Compose, Firebase**.
- **Robust AI** capabilities using Gemini API for advanced scheme understanding and reasoning.
- **Curated, high-quality public datasets** validate core recommendation logic for MVP development.

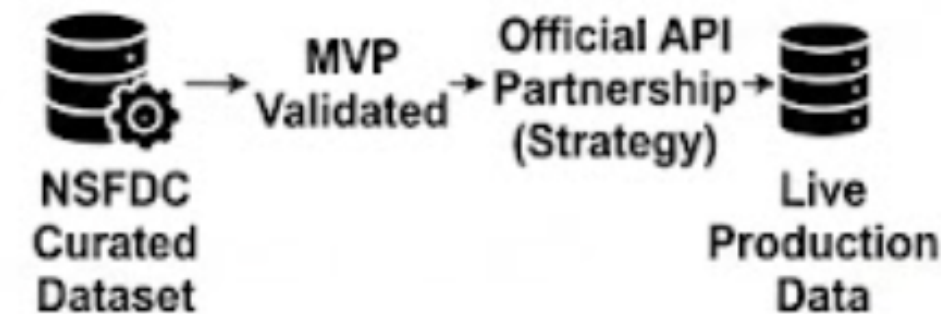
Challenges

- **Data Access:** Direct, real-time Channel Partner fund-utilization or NPA data is currently non-public.
- **Usability & Trust:** Complexity for first-time, non-technical rural users in multi-lingual contexts.
- **Regulatory & Official Data APIs:** Securing institutional partnerships for production-grade live data.

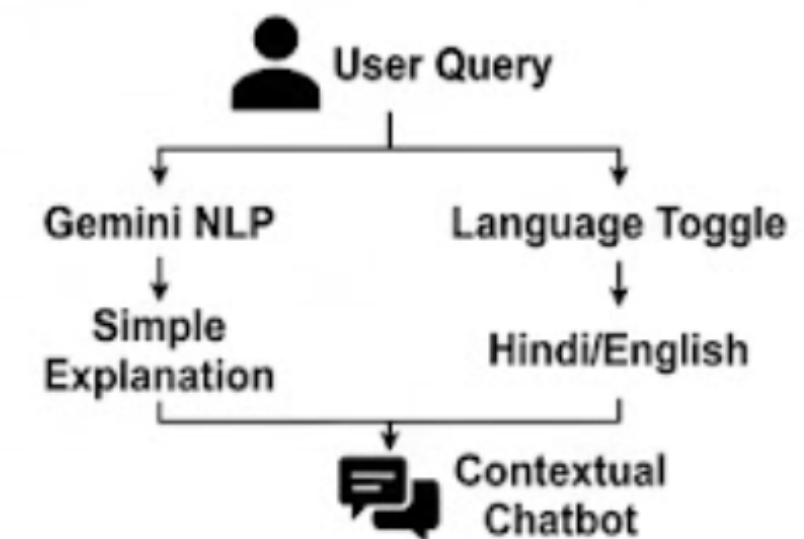
Commercial Feasibility & Strategy

- **Market Validation:** High demand among marginalized entrepreneurs for simplified, accessible, and transparent government scheme discovery.
- **Technical Readiness:** Highly feasible MVP built using curated NSFDC datasets, ready for immediate pilot testing.
- **Scalability:** Designed to integrate directly with official government APIs (NSFDC) for production-level live data streaming.

Solutions & Strategy

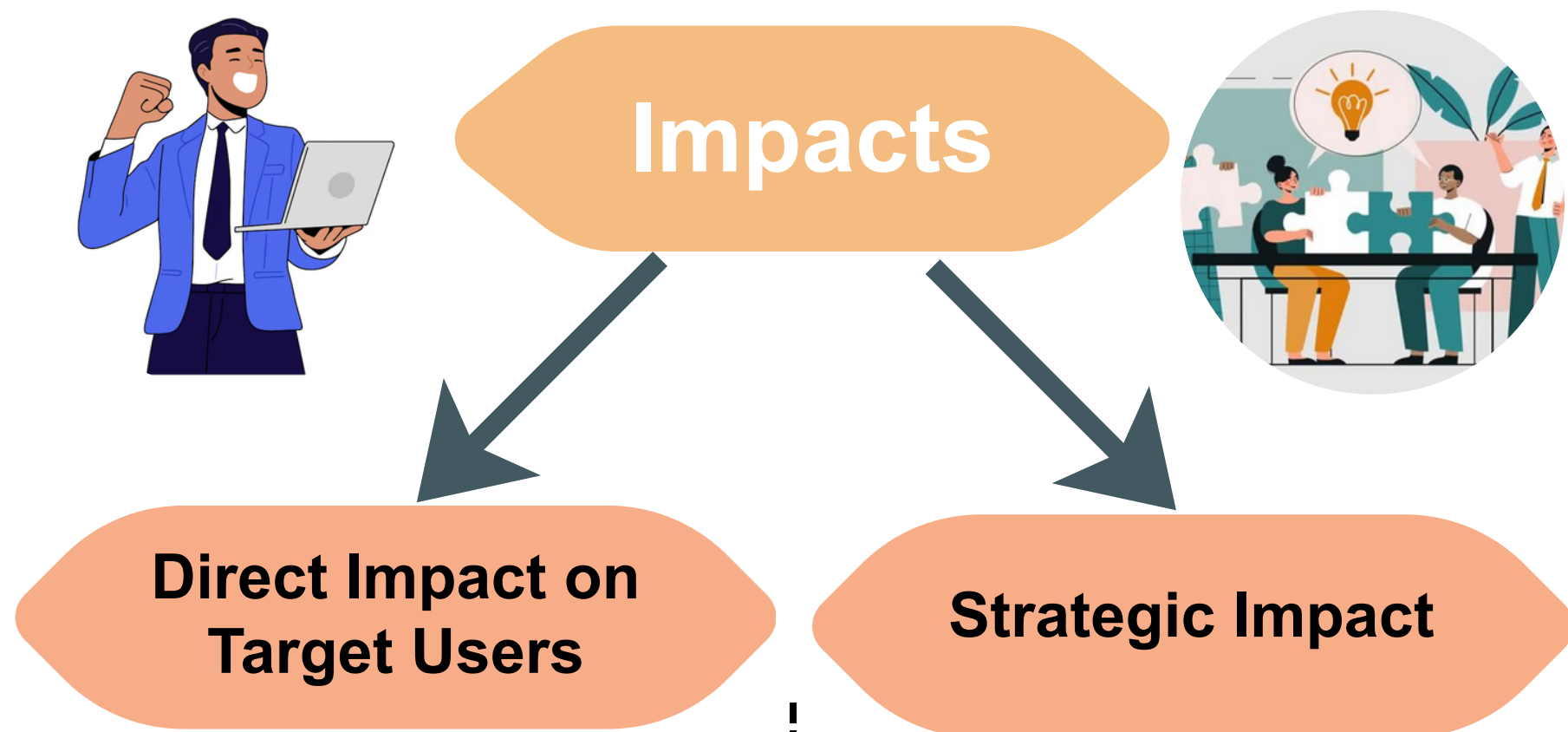


Partnership Strategy for Live API

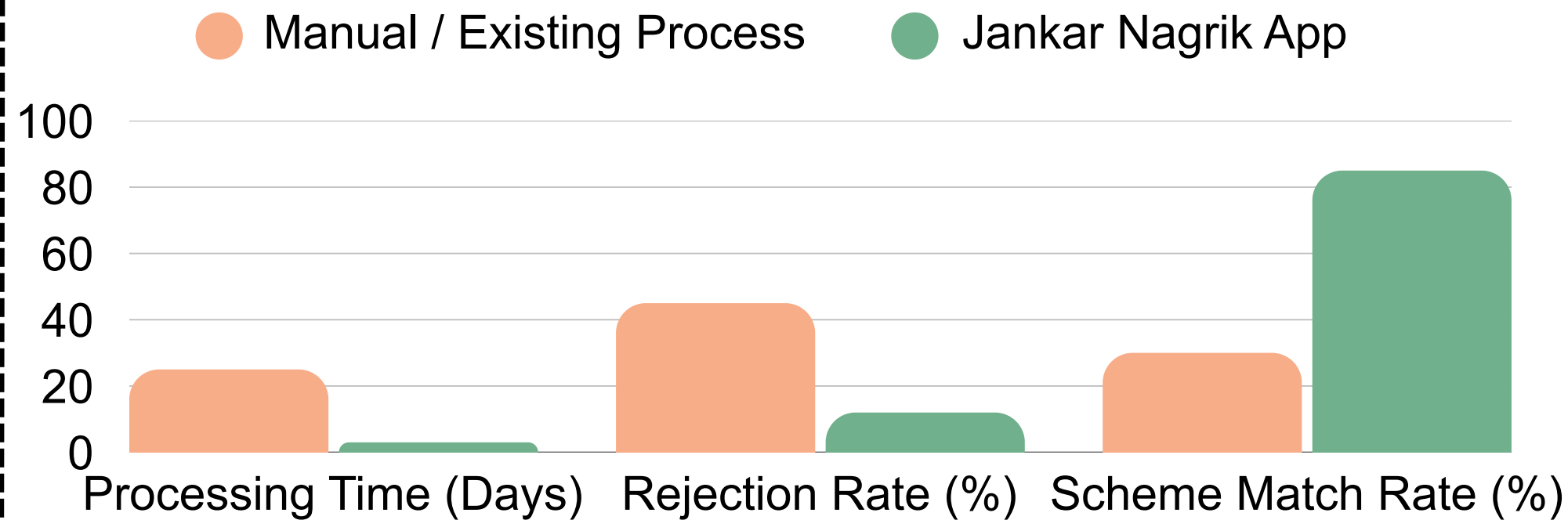
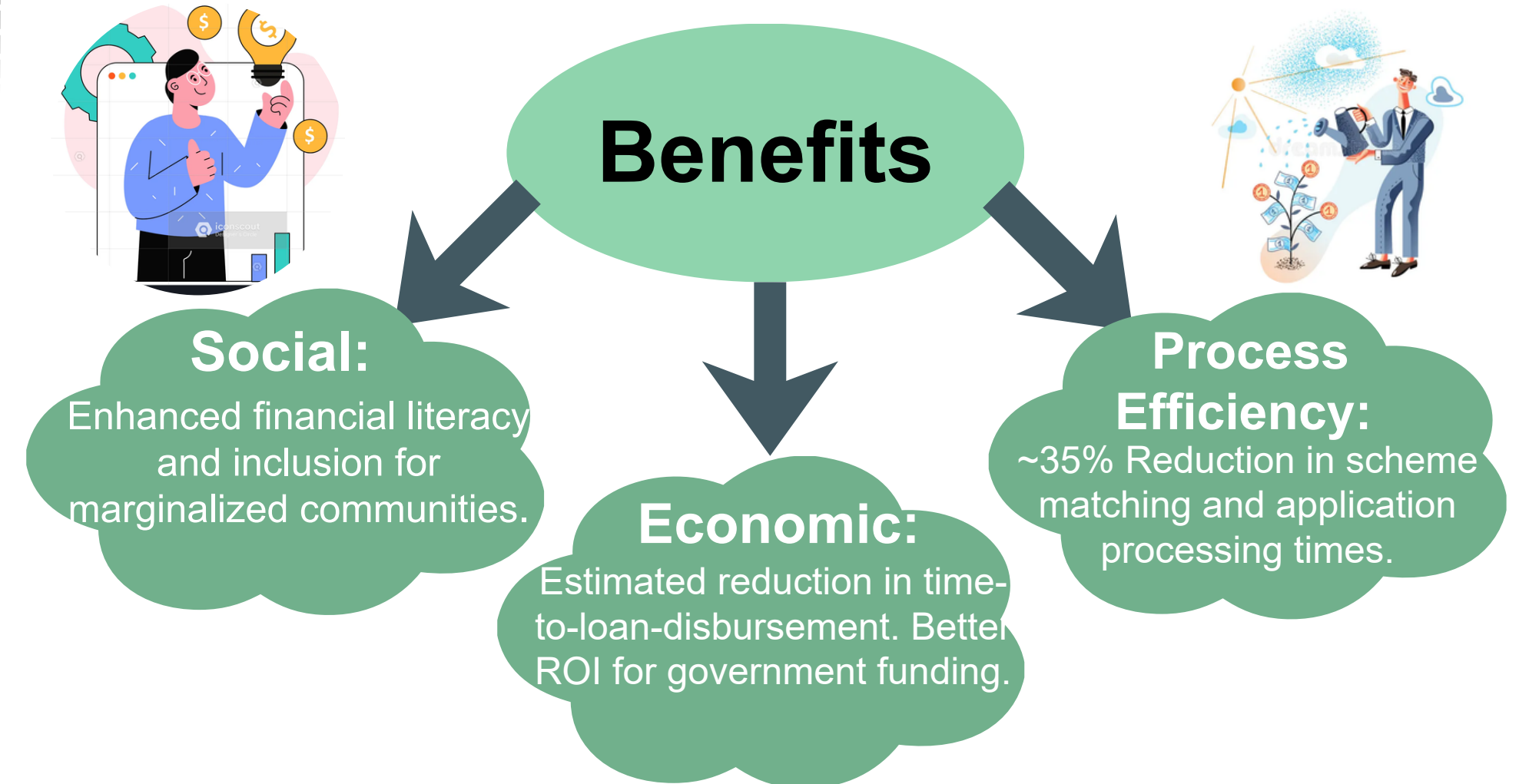


Multi-lingual Gemini Reasoning

IMPACT AND BENEFITS



- **SC Entrepreneurs:** Enables access to concessional loans, reducing rejection rates.
- **Marginalized Micro-Enterprises:** Connects to tailored schemes, overcoming data and language barriers.
- **Partner Organizations (SCA):** Better fund utilization through data-driven matching.
- **Government Governance:** Provides real-time insights into scheme efficacy and regional needs.
- Boosts **grassroots entrepreneurship** for marginalized communities.
- Increases **official financial inclusion** and reduces **informal, high-cost lending**.
- Contributes to '**Aatmanirbhar Bharat**' by making **government support accessible**.



Illustrative projection based on typical manual process inefficiencies; to be validated during pilot testing

NATIONAL PORTAL OF INDIAindia.gov.in

Reference:

**Official government scheme
& service information****myScheme**myscheme.gov.in

Reference:

**Scheme discovery &
eligibility-based matching****NSFDC**nsfdc.nic.in

Reference:

**Loan eligibility, schemes &
Channelising Agencies****PM-SURAJ**pmsuraj.dosje.gov.in

Reference:

**Digital loan application
pathway****GOOGLE GEMINI API**ai.google.dev

Reference:

**AI-assisted contextual
guidance****GOOGLE MAPS PLATFORM**developers.google.com/maps

Reference:

**Location-based Channel
Partner discovery**